



Solar cells using Nanotechnology

glance at the prices at your local gas station will confirm this. The reason behind this is the huge mismatch between the current energy demand and supply – and this gap is only growing wider. The demand for energy is expected to grow as China and India industrializes; at the same time, global oil production is declining, while the climatic effects

of burning fossil fuels have become more pronounced. This brewing crisis will probably lead to more political instability and warfare and it would be fair to say that energy has become a “gun-to-the-head” issue.

The only good that has come out of this energy crisis is that it has spurred scientists to use nanotechnology to find energy alternatives, and to find ways to manufacture everything, including nanoproducts, using less energy. But it is very important to come to grips with the fact that Nanotechnology alone cannot solve the energy problem which is more than Nanotech can encompass. To a world which is still evolving a national energy policy, it is downright difficult to even figure out where Nanotechnology will fit in.



Turning crude oil into diesel fuel

Even though we don't precisely how Nanotechnology fits in, it is easy to visualize the various ways, it can potentially contribute to energy solutions. Many research groups and private companies are already exploring a variety of alternative energy sources, such as solar power, wind power, nuclear fusion, hydrogen fuel cells and biofuels. Slowly but surely new nanotools, knowledge and skills are being gradually absorbed and applied in numerous energy-related technologies, from light bulbs to fuel catalysts to batteries. Some of these advances might occur quietly and unspectacularly, but even small improvements could add up to big savings in energy usage.

To begin a journey toward solutions, the energy problem is divided into well-defined technical challenges. These include efficient energy conversion, efficient energy storage, efficient energy transmission and efficient energy use.